Turbulence Modelling

Application to a turbulent flow downstream of a backward-facing step

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Context & Challenges

Industrial problem

Predicting recirculation and separation zones

Impact on performance (drag), safety (dispersion), efficiency (mixing)

Need for validated turbulence models

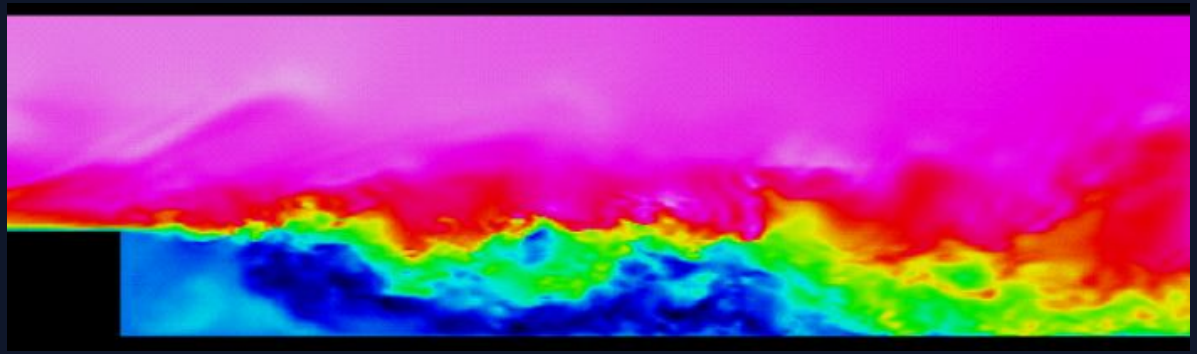
Case study

Simple backward-facing step geometry

Rich physics: separation, recirculation, reattachment

Project Objectives

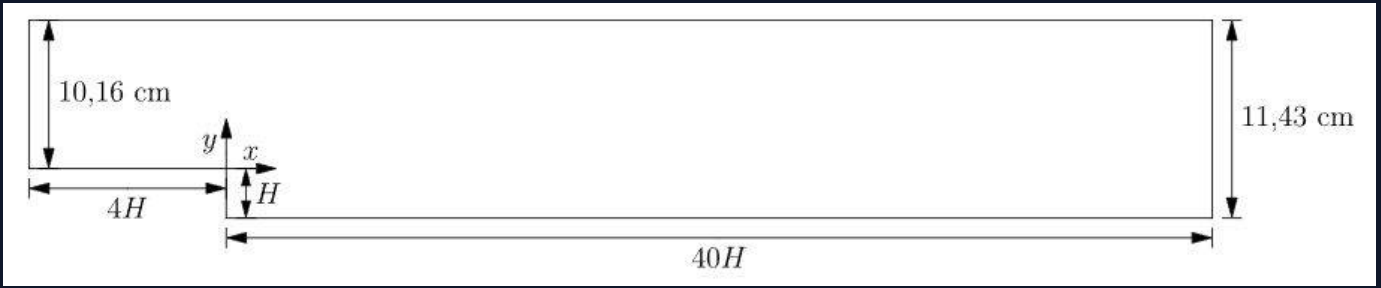
- Apply 2D modelling to a turbulent flow downstream of a step
- 2D RANS simulation with standard k-ε and Realizable k-ε models
- Sensitivity study (mesh, models, boundary conditions)
- Validation against the Driver & Seegmiller (1985) experimental data



Configuration Studied

Geometry

H = 1.27 cm

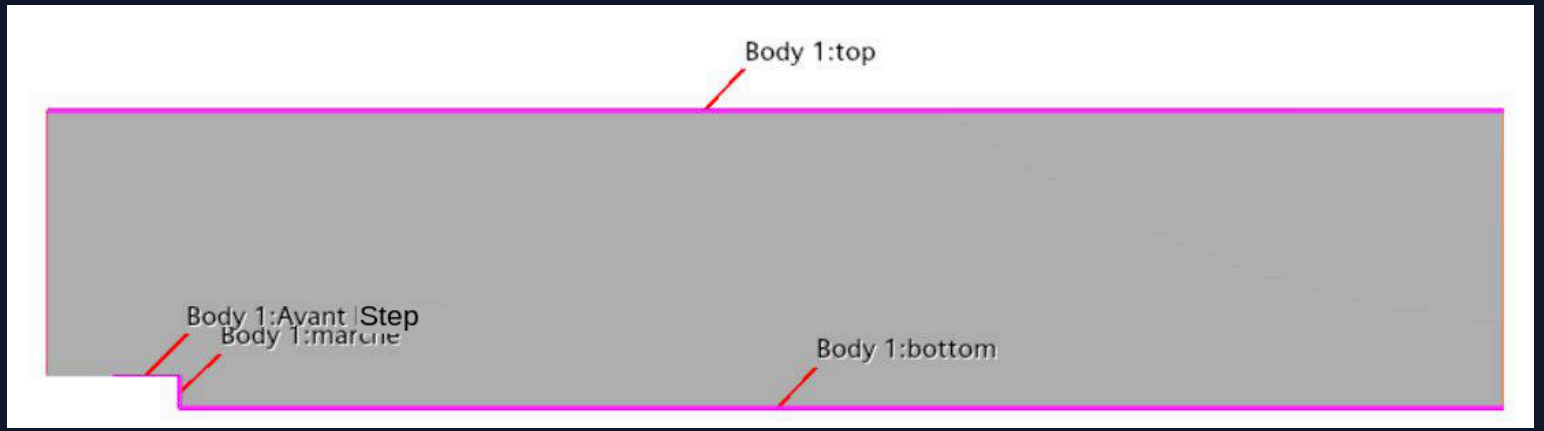


Flow Parameters

- Upstream velocity: U_0 = 38.67 m/s
- Air density: ρ = 1 kg.m-3
- Air kinematic viscosity: ν = 1.5x10-5 m^2.s-1

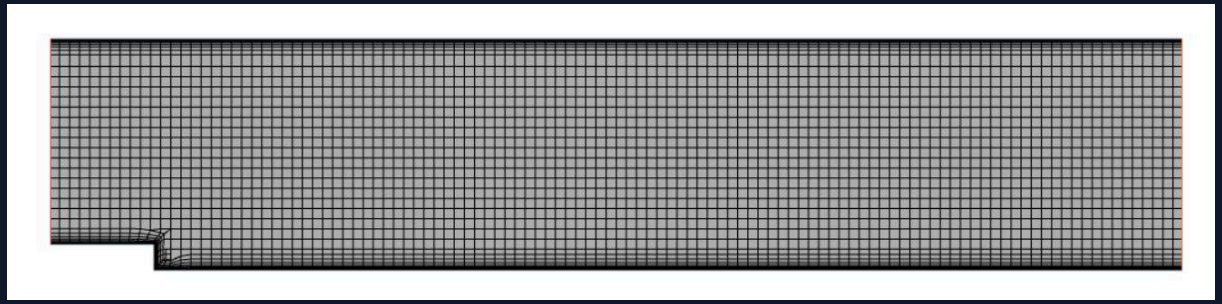
Configuration Studied

Wall names & boundary conditions

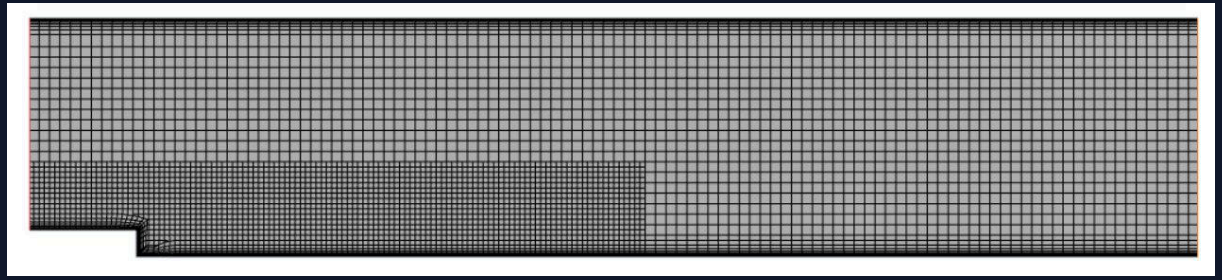


Meshing

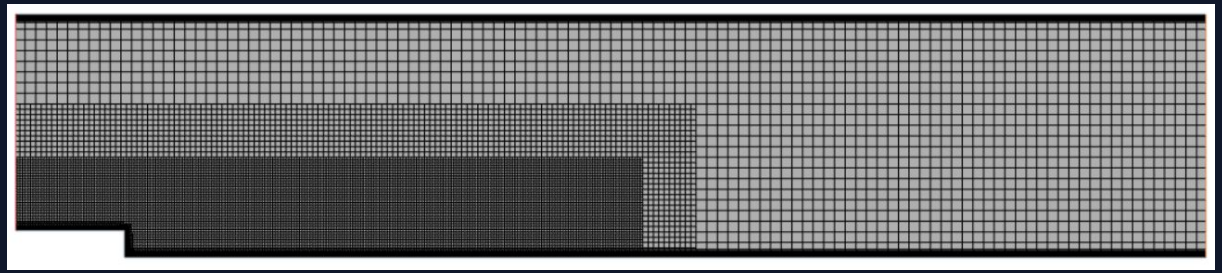
Coarse mesh



Medium mesh

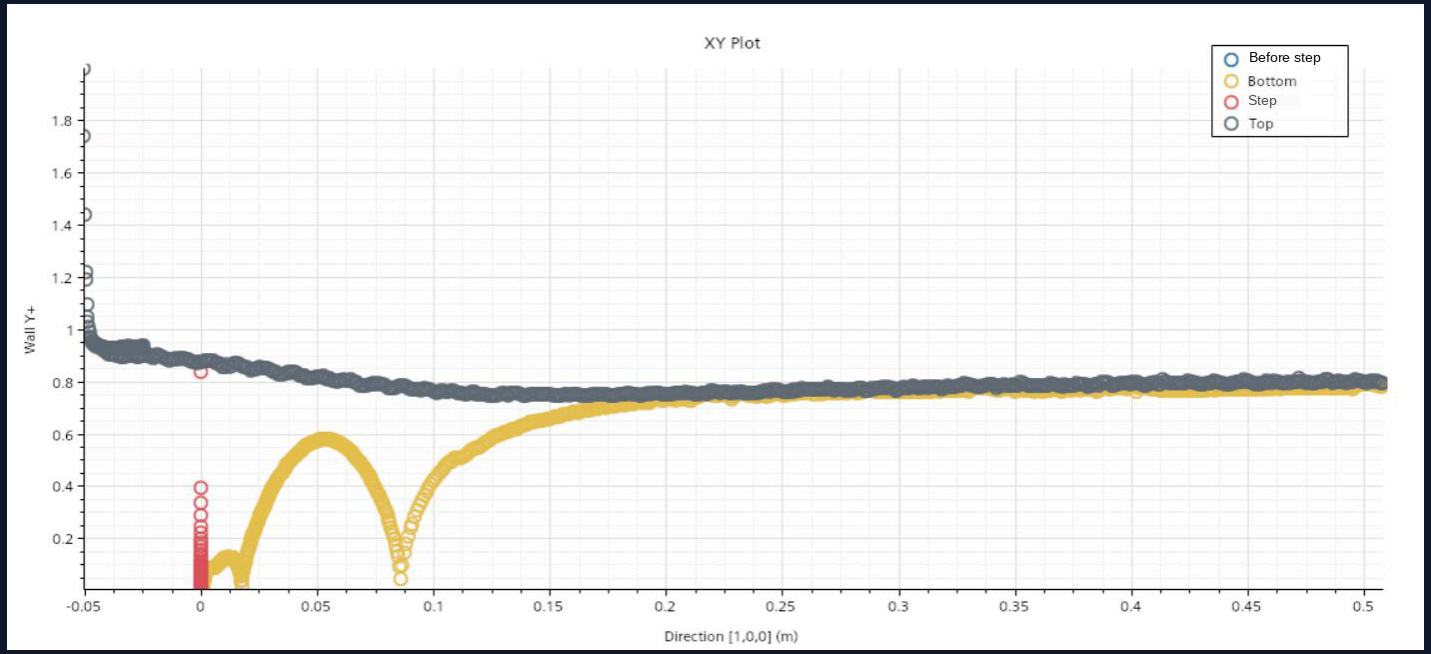


Fine mesh

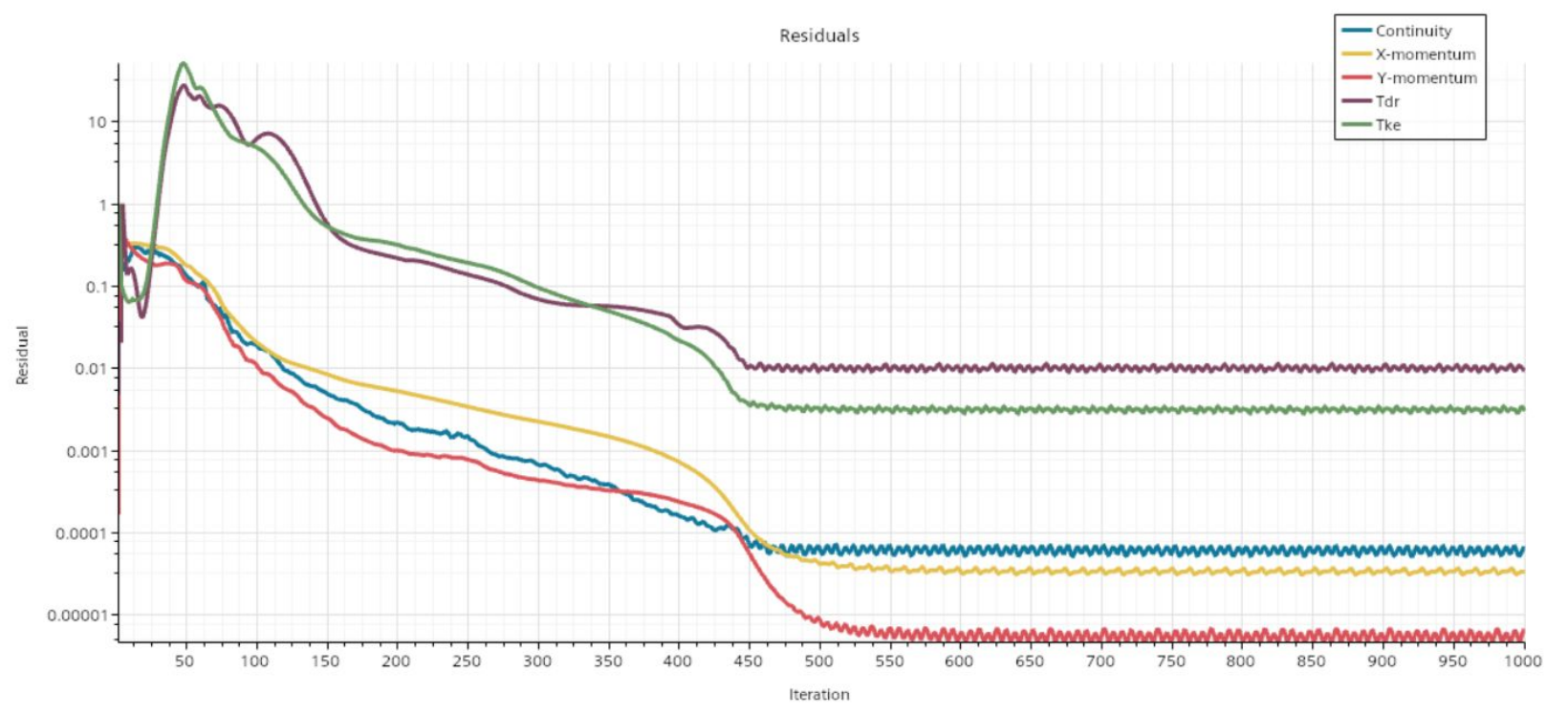


Meshing

Low-Re approach

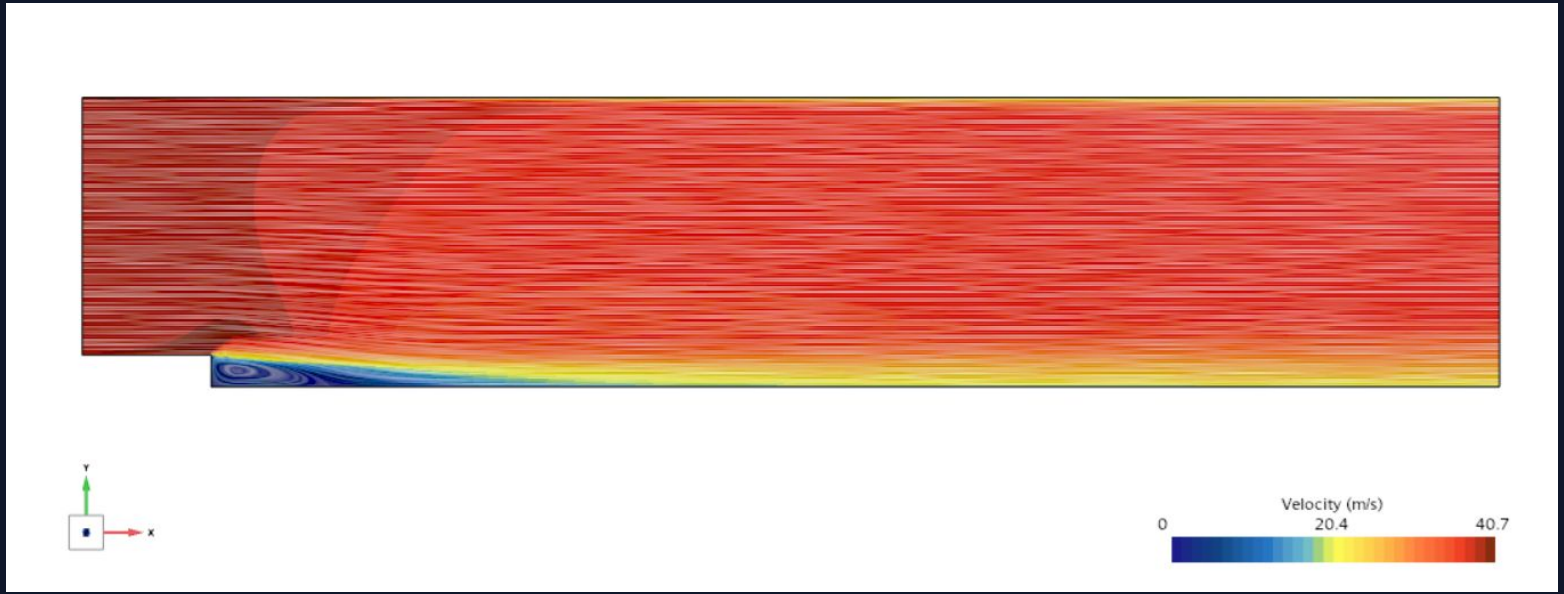


Physical Modelling: Standard k-ε, Low Reynolds

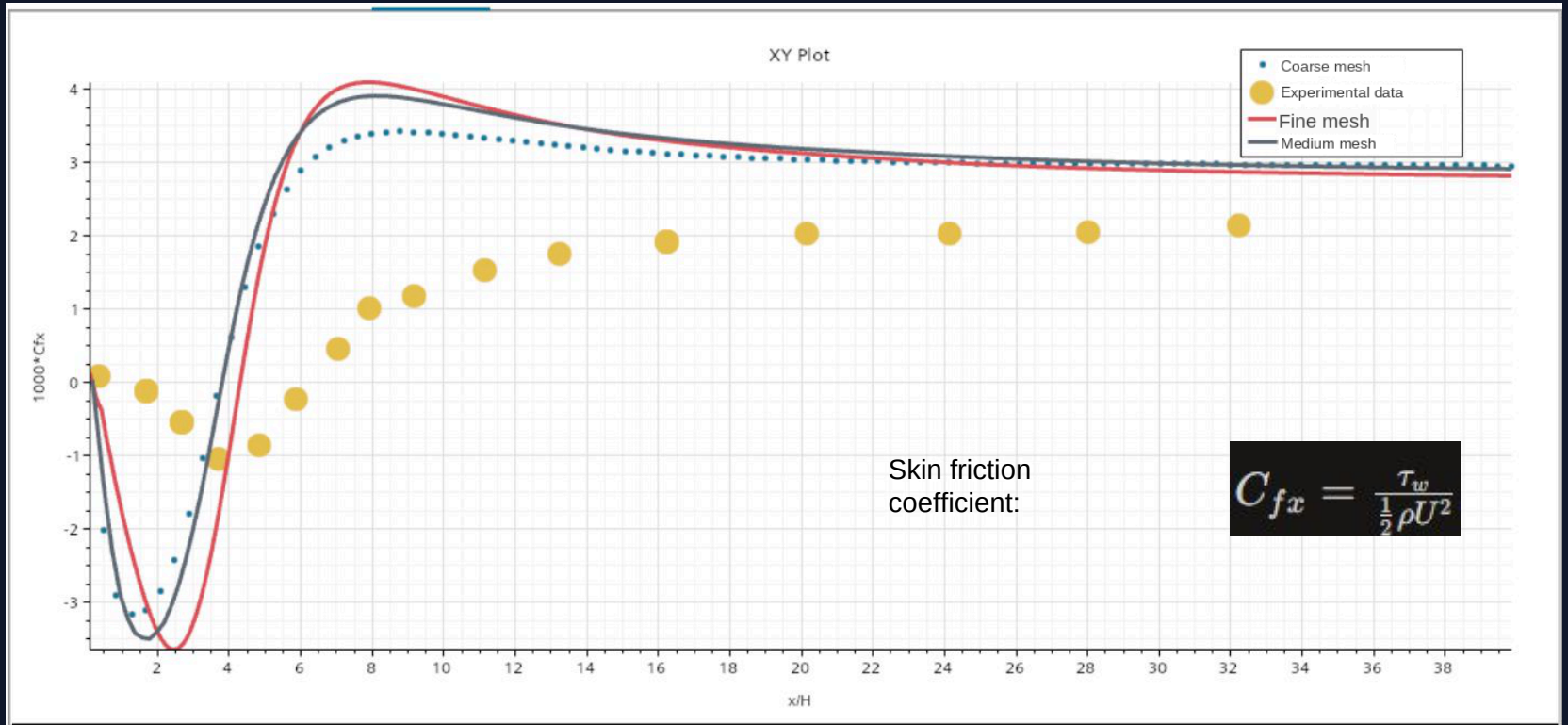


Physical Modelling: Standard k-ε, Low Reynolds

Velocity field



Physical Modelling: Standard k-ε, Low Reynolds



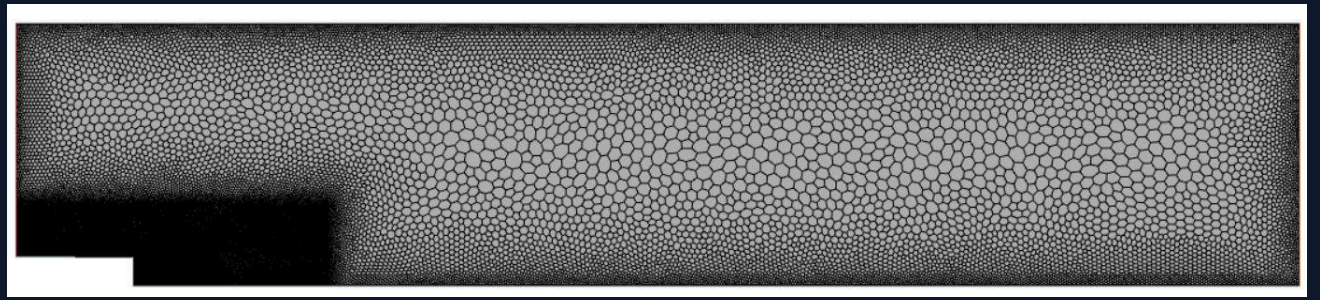
Skin friction coefficient:

$$C_{fx} = \frac{\tau_w}{\frac{1}{2}\rho U^2}$$

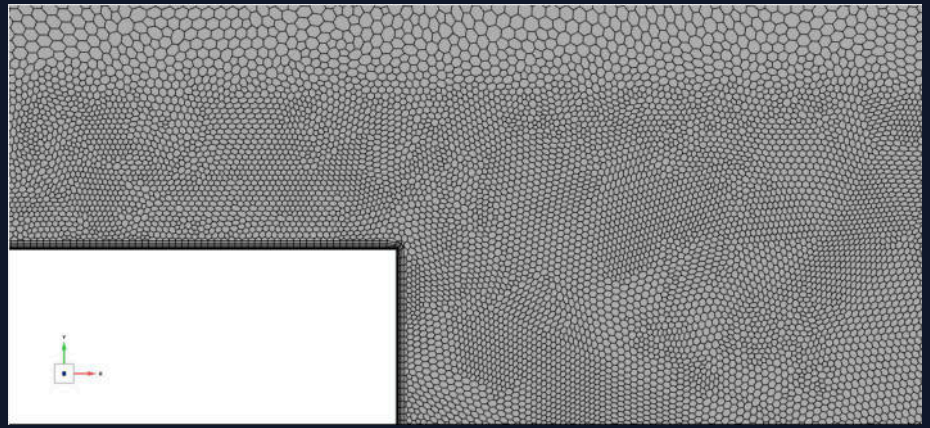
Final Mesh

- Base size: 0.001 m
- Refinement zone:

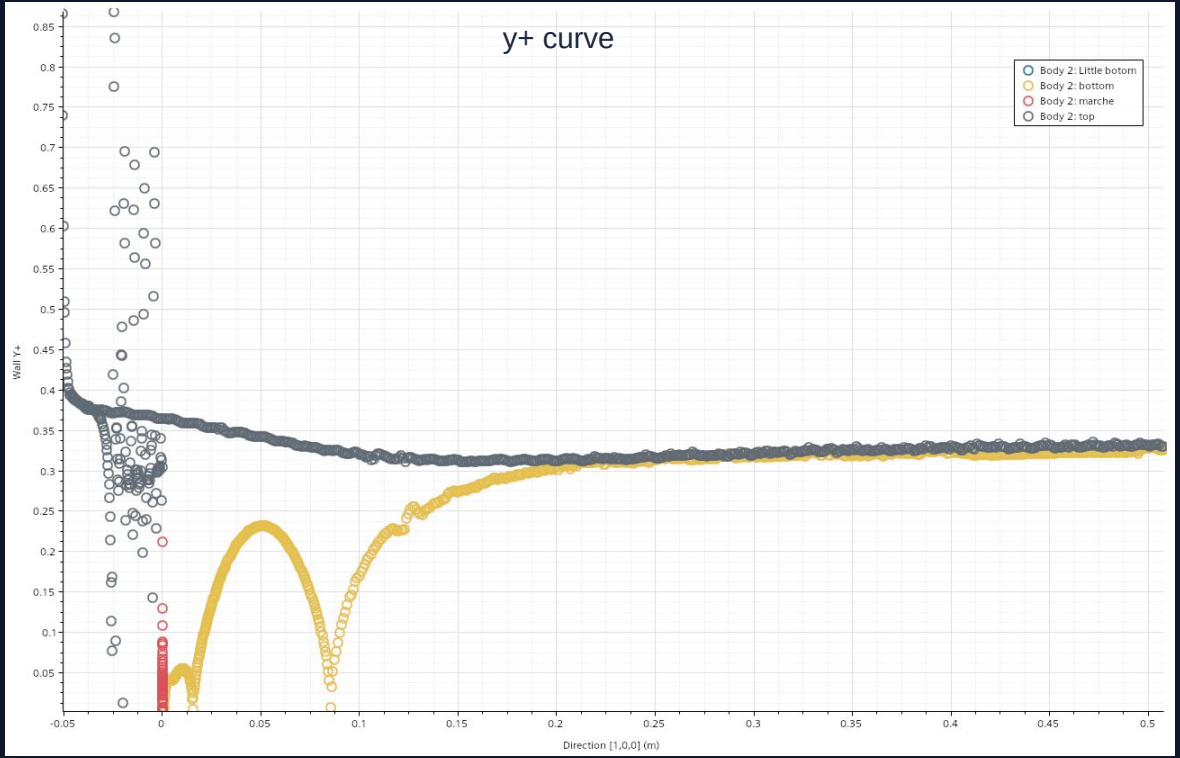
  0.5 BS and 0.33 BS
- Prism layers: 10
- Total thickness: 33% of base
- Stretching: 1.3



Polyhedral mesh generated directly in 2D (2D meshing badge) to modify the mesh faster without going back through 3D



Validating the Low-Re Approach



Influence of Boundary Conditions

Mean velocity profile

$$u(y) = U_{CL}\left(\frac{y}{D}\right)^{1/7} \text{ pour } 0 < y < D$$

Where:

- UCL = 8U0/7 (m/s)
- y, the distance to the wall
- D, the hydraulic diameter



Turbulent kinetic energy k Dissipation rate ε

$$k = \frac{3}{2}(U_0 I)^2$$

With turbulence intensity I:

$$I = \frac{u'}{U_0} = 0,16(Re_{D_H})^{-1/8}$$

And the flow Reynolds number ReDH:

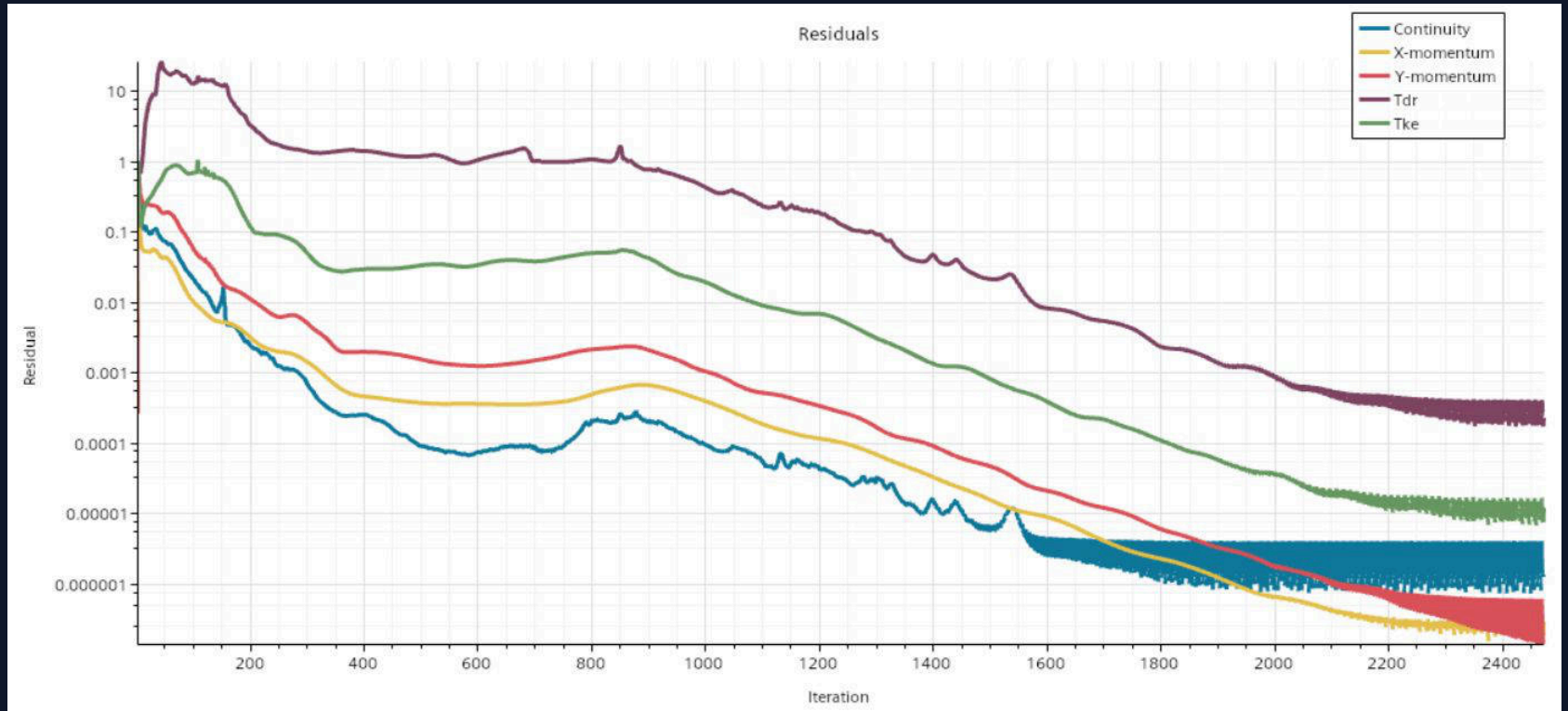
$$Re_{D_H} = \frac{U_0 D_H}{\nu}$$

$$\epsilon = \frac{C_\mu^{3/4} k^{3/2}}{l}$$

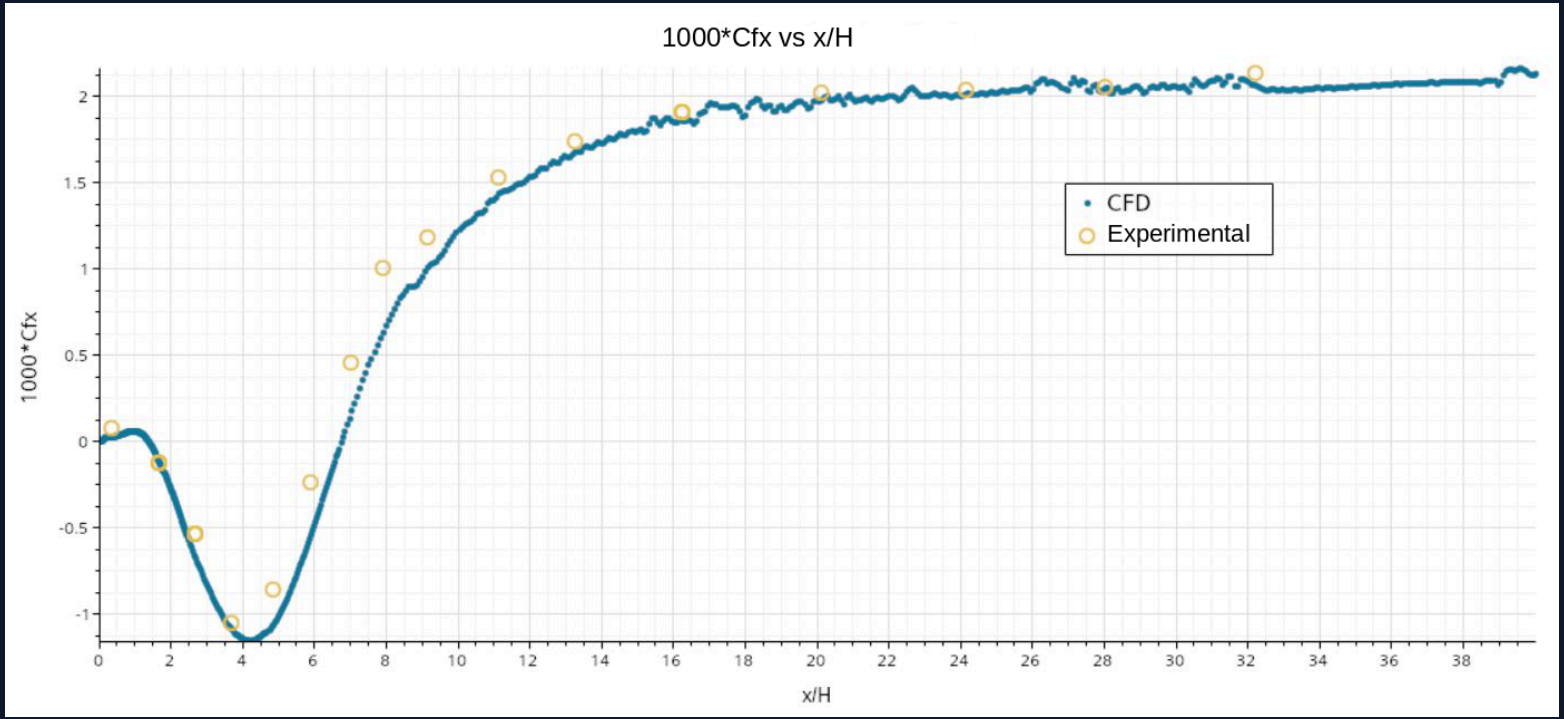
With:

- Cμ = 0.09
- And the turbulent length scale: l = 0.07 * DH

Physical Modelling: Realizable k-ε, Low Reynolds



Physical Modelling: Realizable k-ε, Low Reynolds



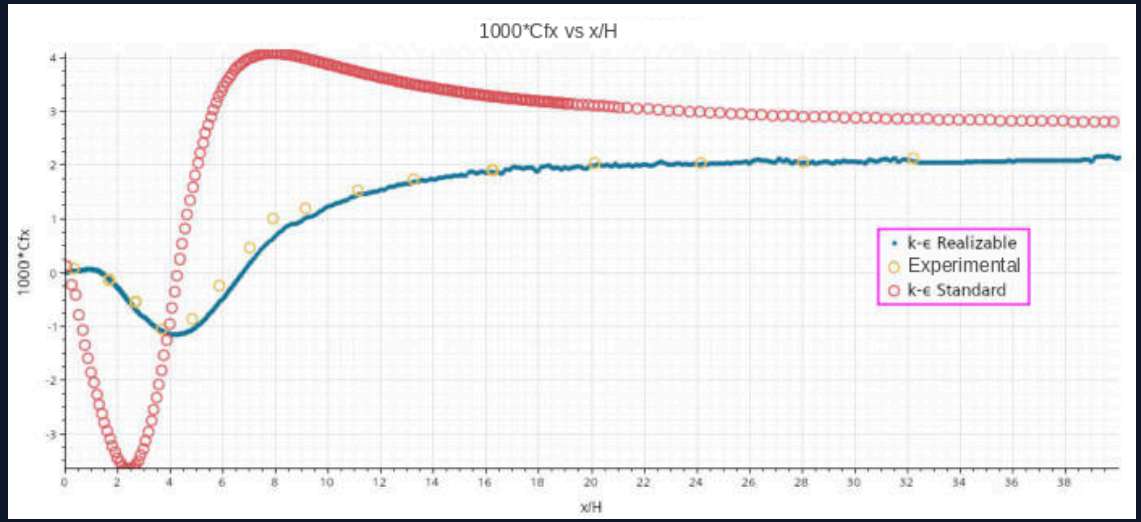
Comparing the Two k-ε Models

k-ε Standard

Residuals ≈ 10-2 - 10-5
Recirculation zone error: 33%

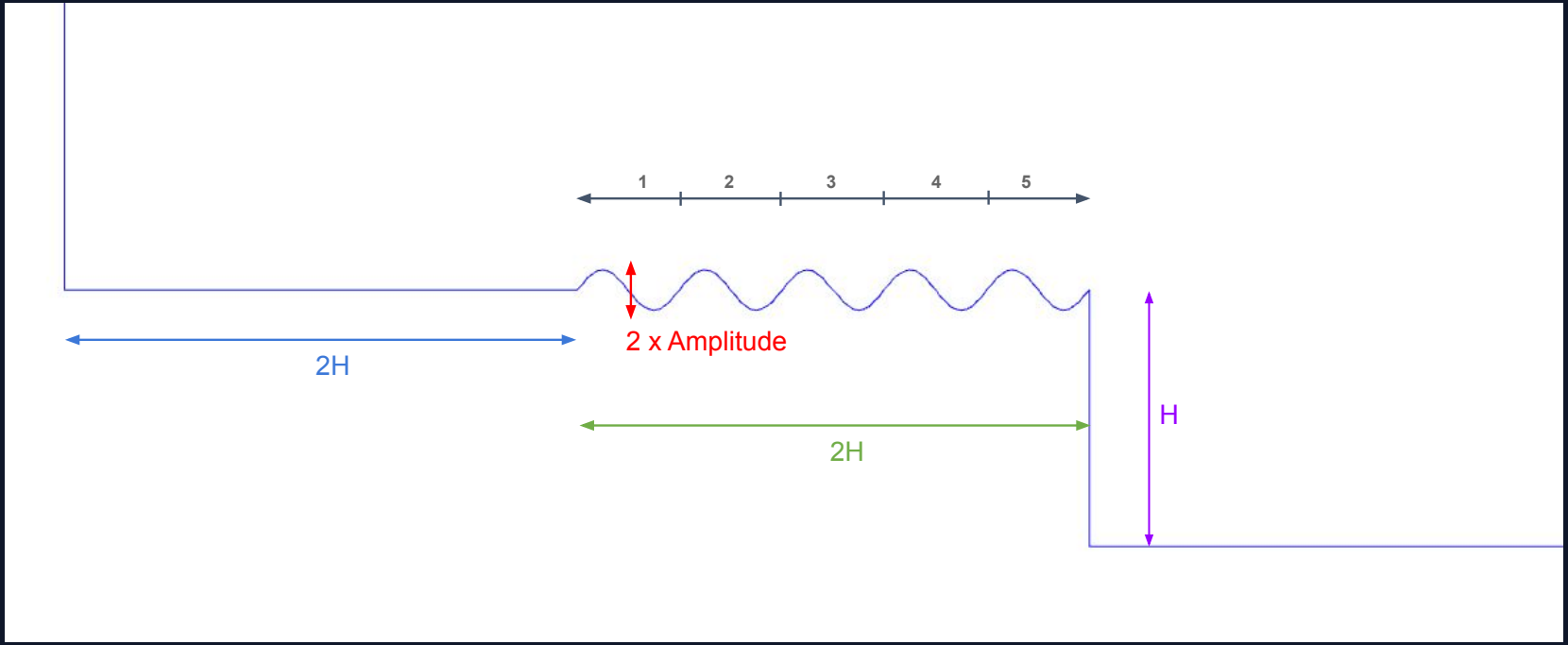
k-ε Realizable

Residuals ≈ 10-3 - 10-6
Recirculation zone error: 7.45%



Controlling the Recirculation Zone

Passive Control - Sinusoidal Wall Concept



Passive Control - Geometry

X	Y	Z
0.0000	0.0000	0.0000
0.0000	-0.0127	0.0000
0.5080	-0.0127	0.0000
0.5080	0.1016	0.0000
-0.0508	0.1016	0.0000
-0.0508	0.0000	0.0000
-0.0254	0.0000	0.0000
Coordonnee sinus		
0.0000	0.0000	0.0000

Nombre de sinus	5
Resolution	20
Nombre de point X	100
Emplitude Sinus	0.0005

Coordonnee Y pour une periode	
Point	Coordonnee
1	0.0002
2	0.0003
3	0.0004
4	0.0005
5	0.0005
6	0.0005
7	0.0004
8	0.0003
9	0.0002
10	0.0000
11	-0.0002
12	-0.0003
13	-0.0004
14	-0.0005
15	-0.0005
16	-0.0005
17	-0.0004
18	-0.0003
19	-0.0002
20	0.0000

Coordonnee sur X	Numero du point	Coordonnee sur Y
-0.0251	1	0.000154508497187474000000000000
-0.0249	2	0.000293892626146237000000000000
-0.0246	3	0.000404508497187474000000000000
-0.0244	4	0.000475528258147577000000000000
-0.0241	5	0.000500000000000000000000000000
-0.0239	6	0.000475528258147577000000000000
-0.0236	7	0.000404508497187474000000000000
-0.0234	8	0.000293892626146237000000000000
-0.0231	9	0.000154508497187474000000000000
-0.0229	10	0.000000000000000000061232339957
-0.0226	11	-0.000154508497187474000000000000
-0.0224	12	-0.000293892626146236000000000000
-0.0221	13	-0.000404508497187474000000000000
-0.0218	14	-0.000475528258147577000000000000
-0.0216	15	-0.000500000000000000000000000000
-0.0213	16	-0.000475528258147577000000000000
-0.0211	17	-0.000404508497187474000000000000
-0.0208	18	-0.000293892626146237000000000000
-0.0206	19	-0.000154508497187474000000000000
-0.0203	20	-0.000000000000000000122464679915
-0.0201	21	0.000154508497187474000000000000
-0.0198	22	0.000293892626146236000000000000
-0.0196	23	0.000404508497187474000000000000
-0.0193	24	0.000475528258147577000000000000
-0.0191	25	0.000500000000000000000000000000
-0.0188	26	0.000475528258147577000000000000
-0.0185	27	0.000404508497187474000000000000
-0.0183	28	0.000293892626146237000000000000
-0.0180	29	0.000154508497187474000000000000
-0.0178	30	0.000000000000000000183697019872

In the end, 5 different geometries:

- Amplitude 0.0001 m



- Amplitude 0.0002 m



- Amplitude 0.0003 m



- Amplitude 0.0004 m



- Amplitude 0.0005 m



Passive Control - Geometry

Full geometry:



“Background”: “Sinusoid”:



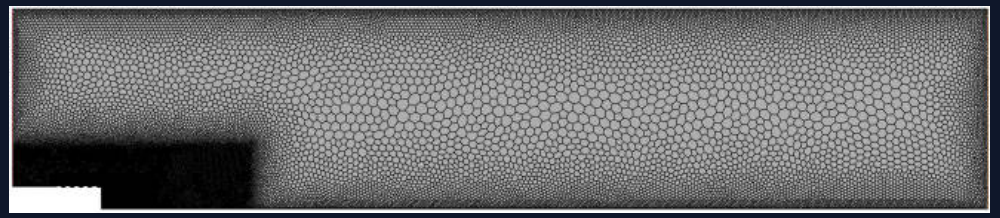


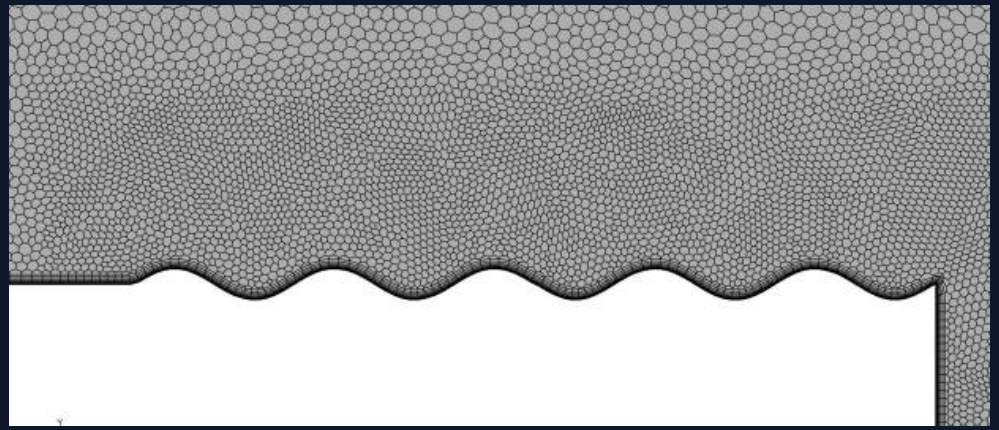
Passive Control - Mesh

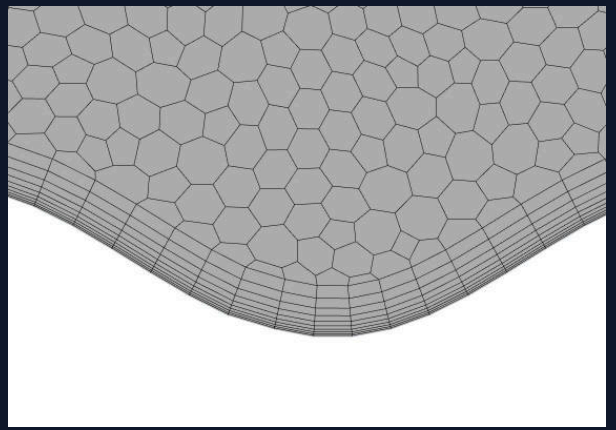
Prism layers: 10

Total thickness: 33% of base

Stretching: 1.3







Passive Control - Simulations

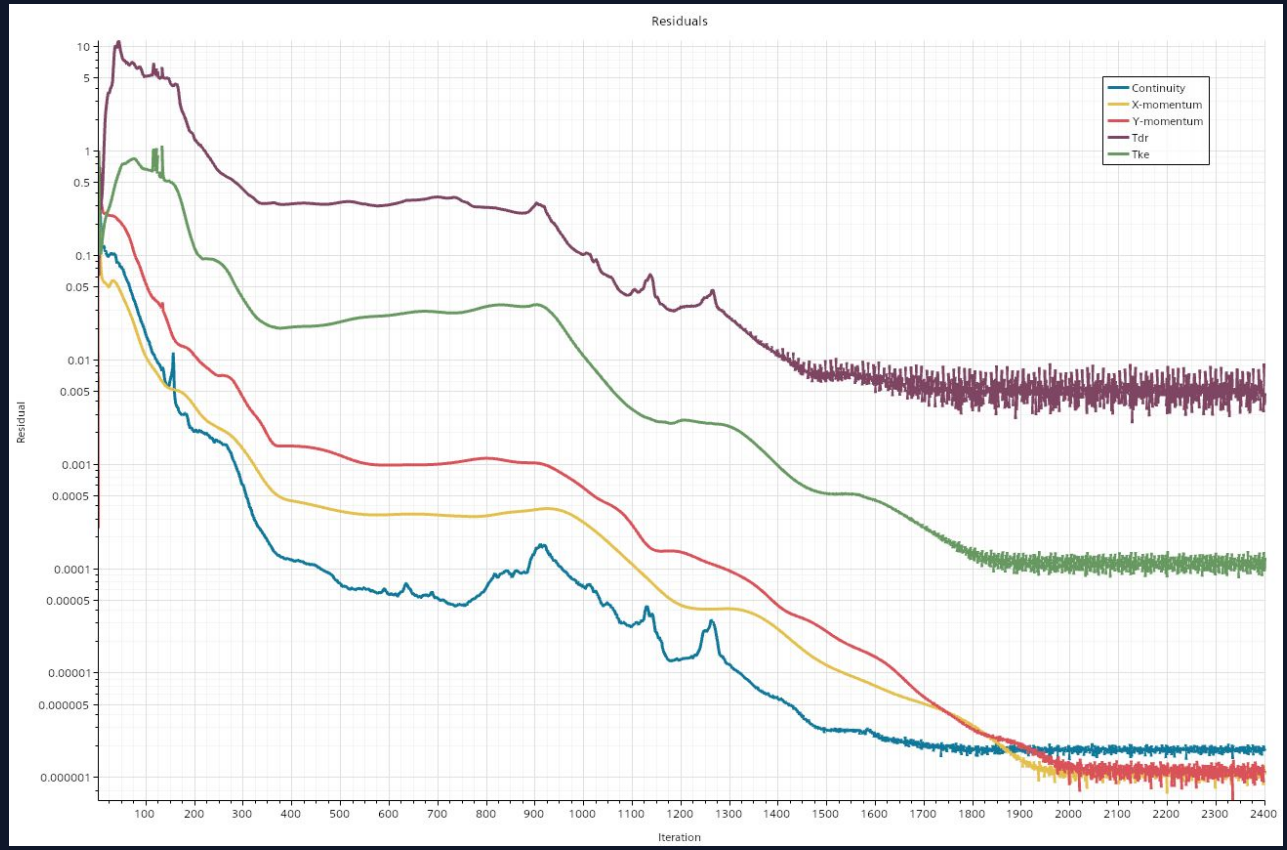
Continuity : 10^-6

X-momentum: 10^-6

Y-momentum : 10^-6

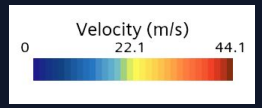
Tdr : 10^-3

Tke : 10^-4

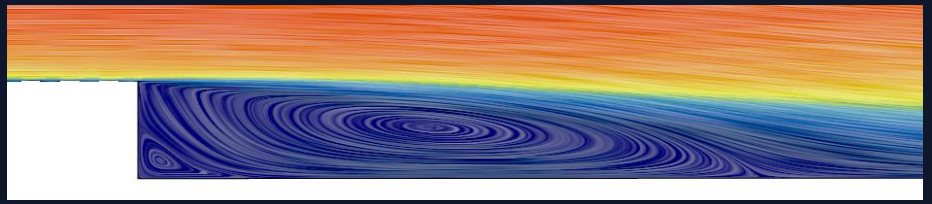


Passive Control - Results

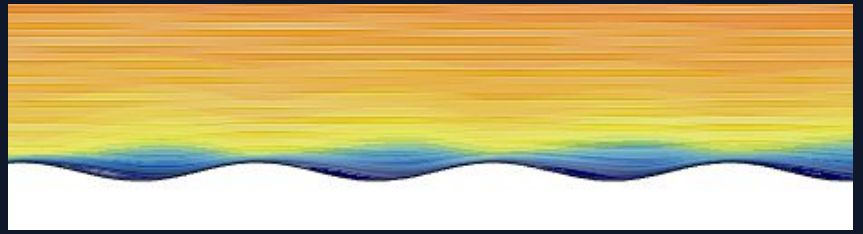
Qualitative results:



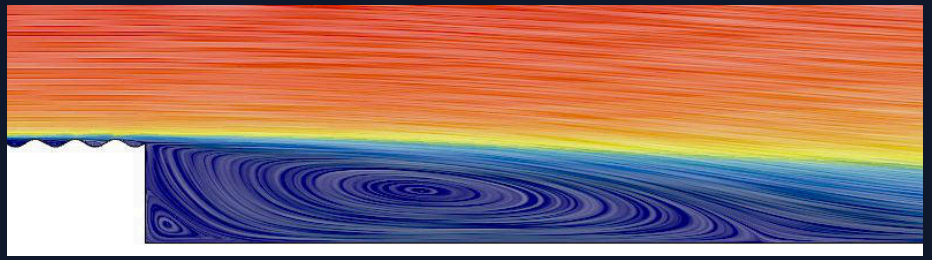
Recirculation zone, amplitude 0.0001 m:



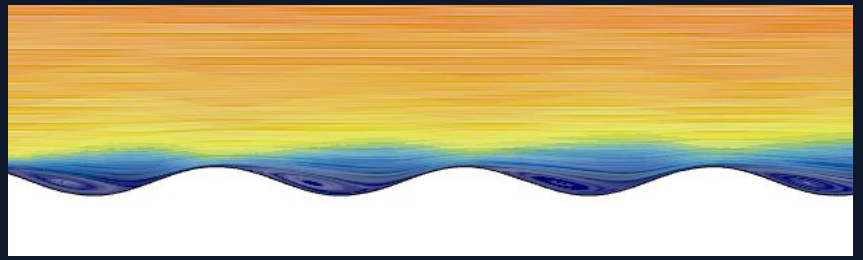
Sinusoid zone, amplitude 0.0002 m:



Recirculation zone, amplitude 0.0005 m:

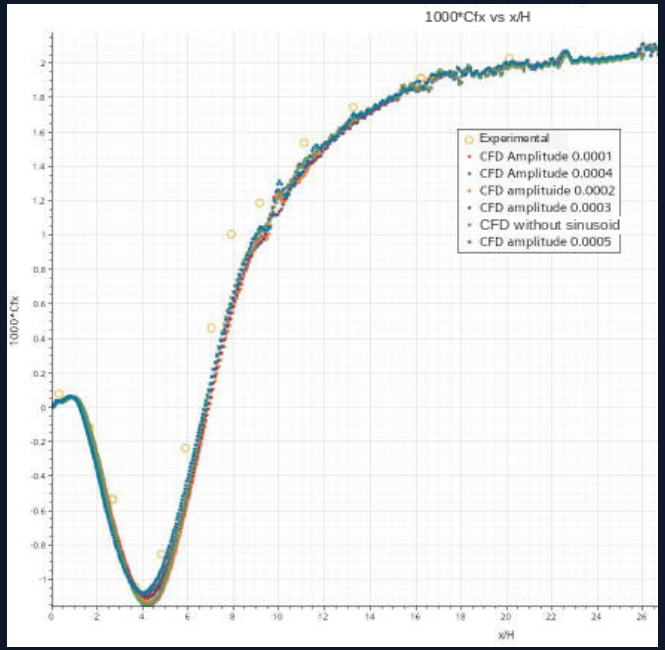


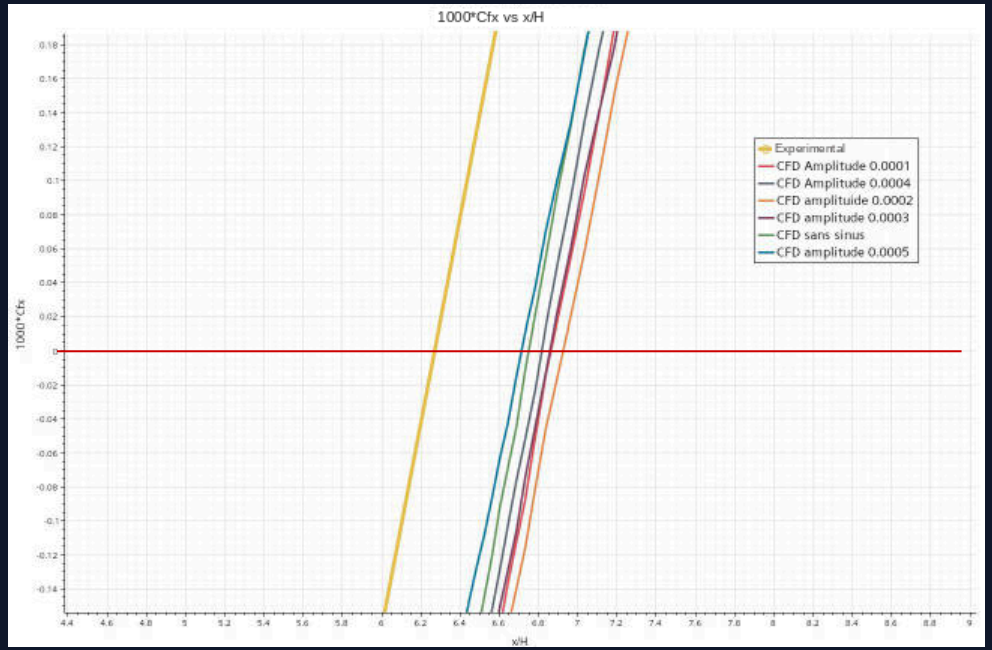
Sinusoid zone, amplitude 0.0003 m:



Passive Control - Results

Quantitative results:



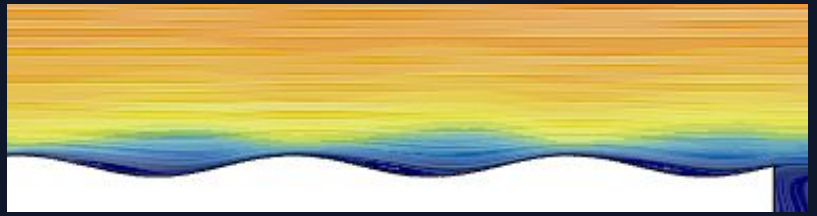


Passive Control - Results

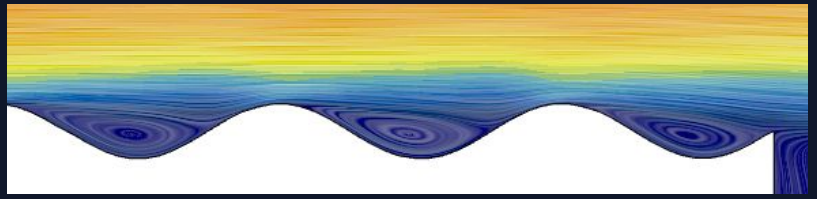
Quantitative results:

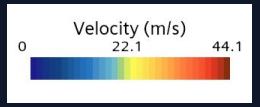
Case	Recirc. zone length	Dev. vs CFD w/o sinusoid
Amplitude 0.0001 m	6.85 m	1.4%
Amplitude 0.0002 m	6.95 m	2.8%
Amplitude 0.0003 m	6.85 m	1.4%
Amplitude 0.0004 m	6.82 m	1.02%
Amplitude 0.0005 m	6.7 m	-0.75%

Velocity vectors, amplitude 0.0002 m:

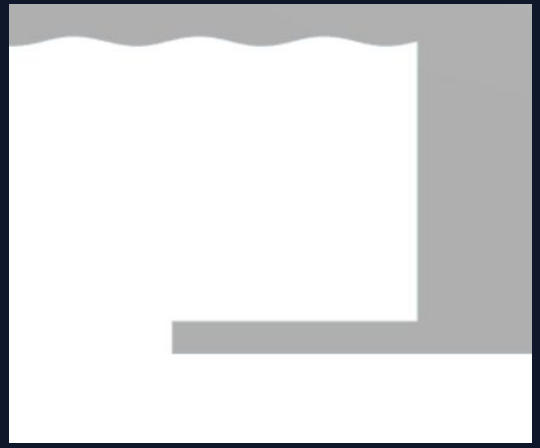


Velocity vectors, amplitude 0.0005 m:

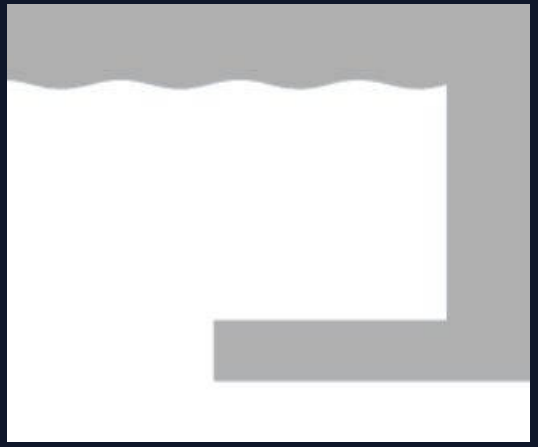




Active Control - Jet Concept



Jet height:
H/10
(0.127 cm)



Jet height:
2H/10
(0.254 cm)

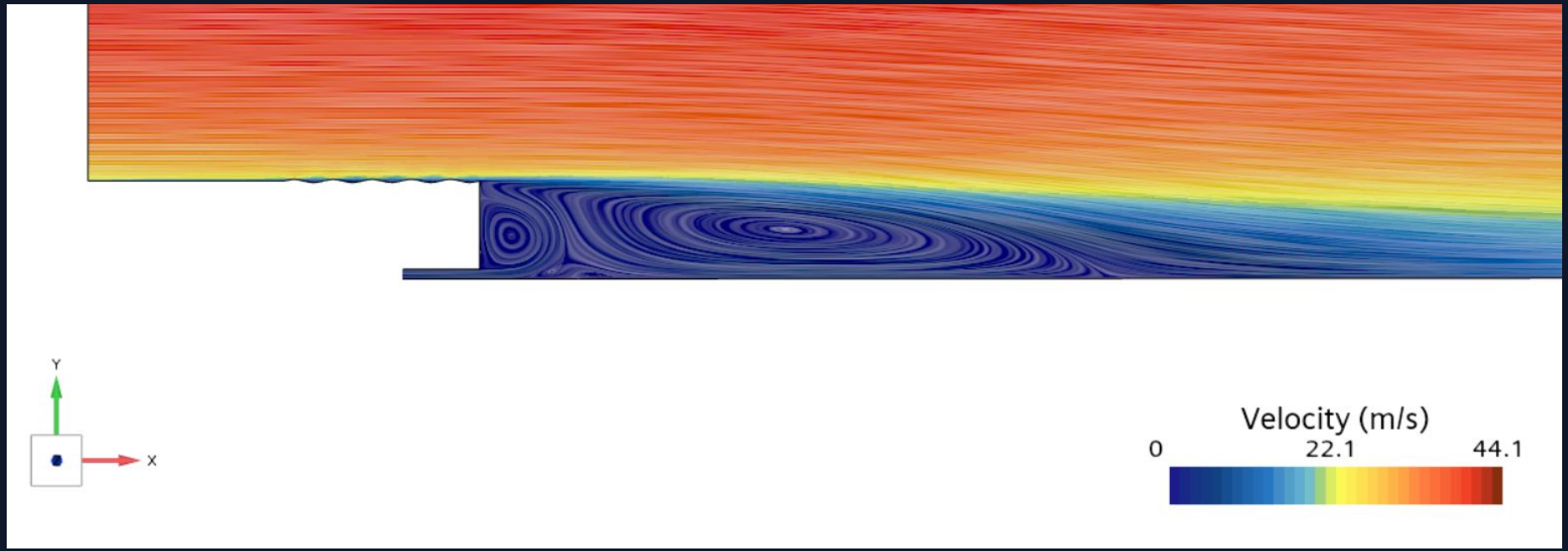


Jet height:
3H/10
(0.381 cm)

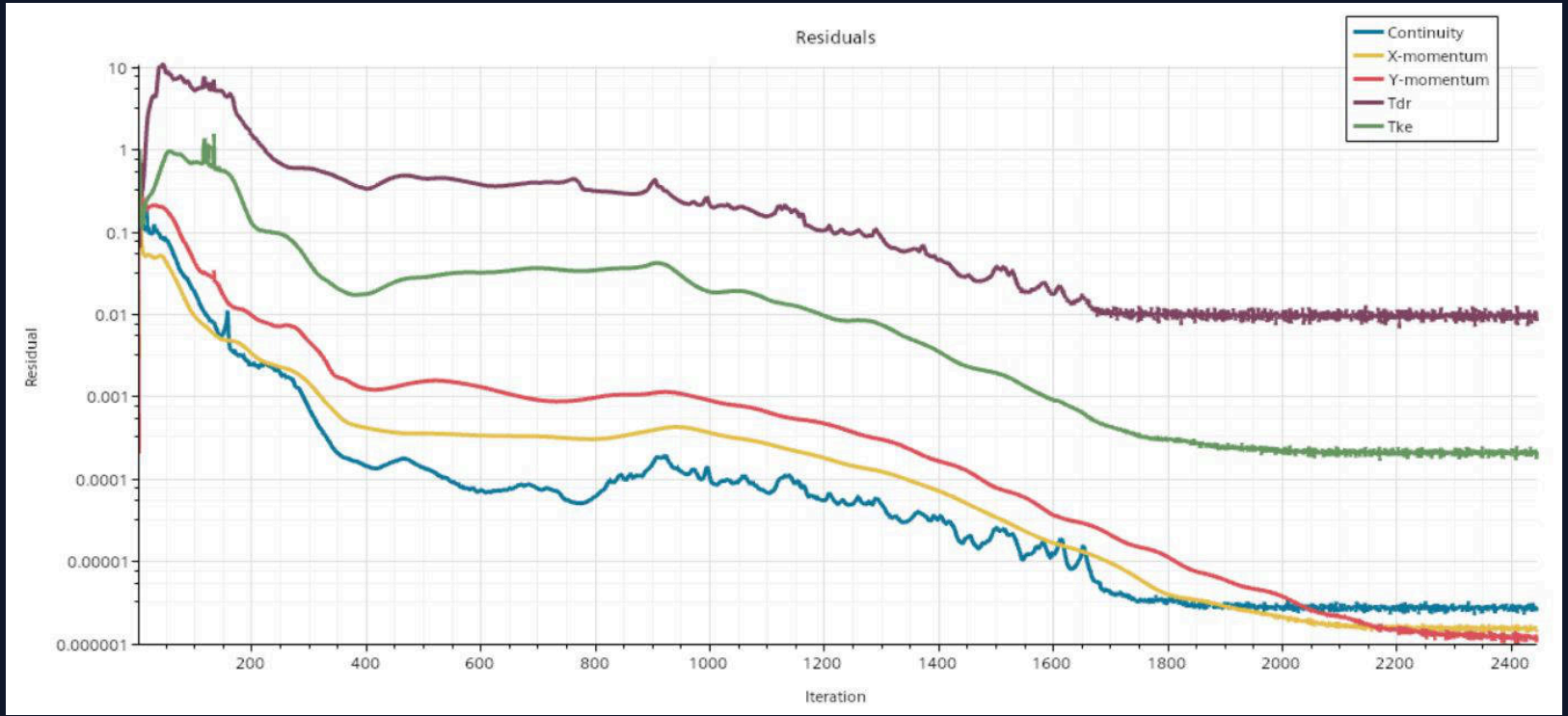
Jet velocity: 3.867 m/s (10% of U_o)

Jet - H/10 Case

Observation: two recirculation zones

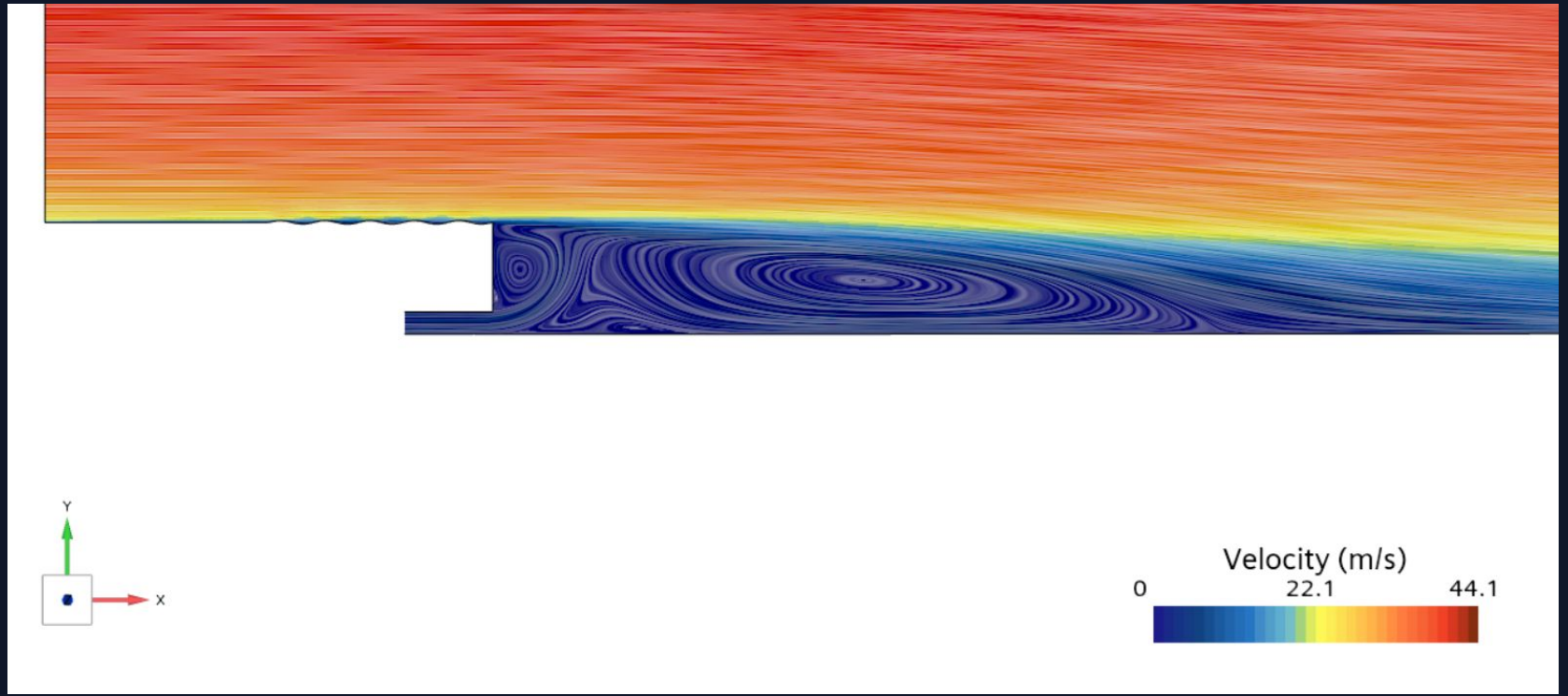


Jet - H/10 Case - Residuals



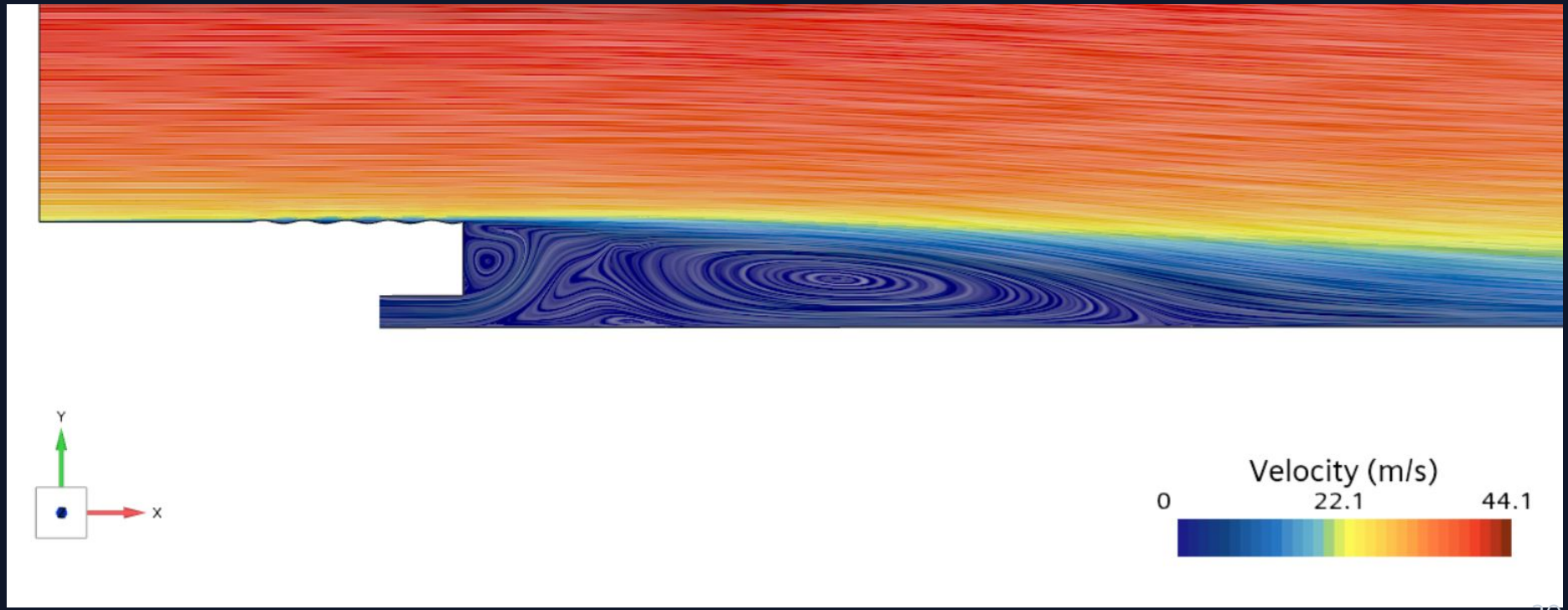
Jet - 2H/10 Case

Observation: two recirculation zones

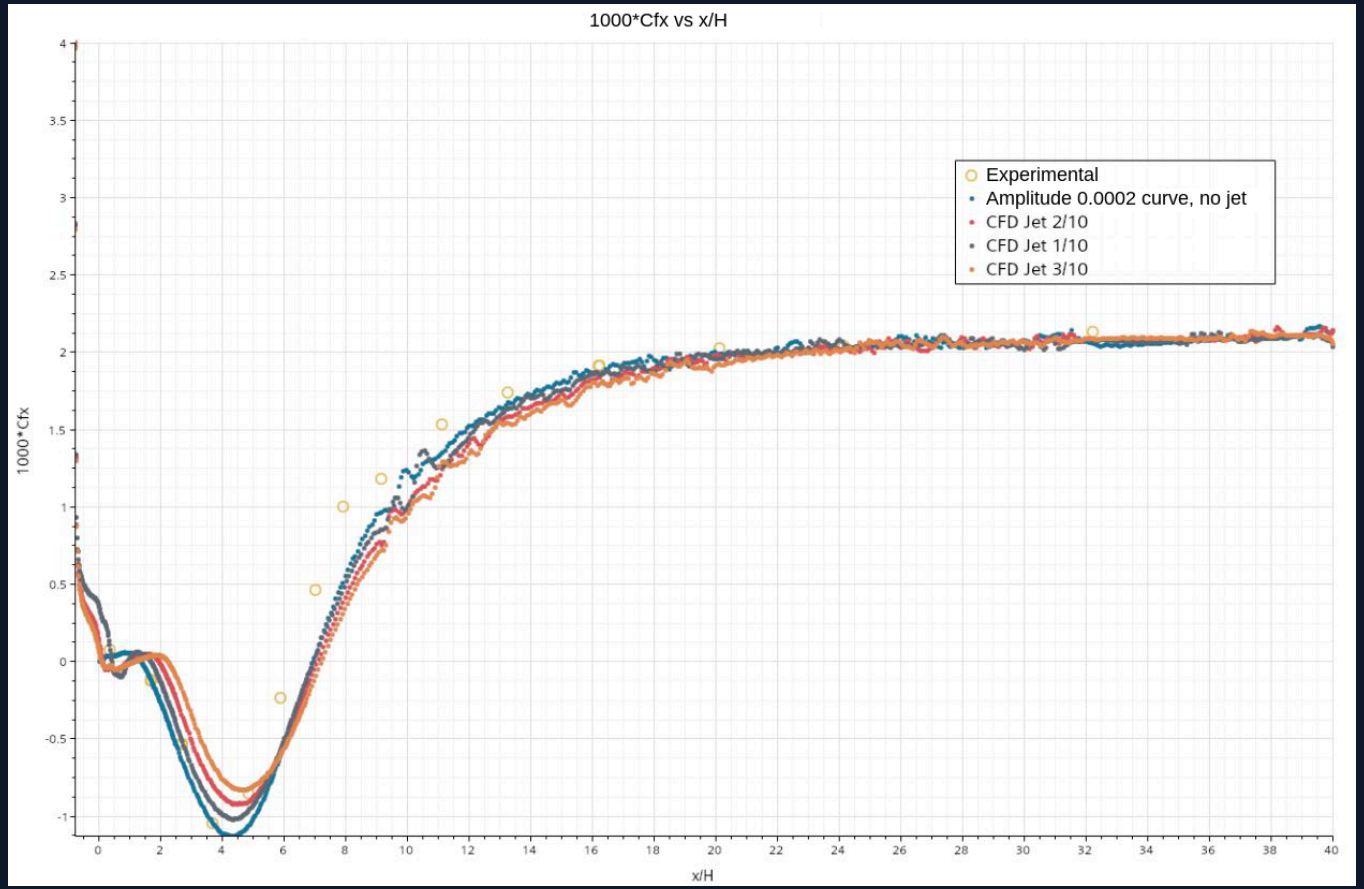


Jet - 3H/10 Case

Observation: two recirculation zones



Jet - Comparing the Configurations



Jet - Comparing the Configurations

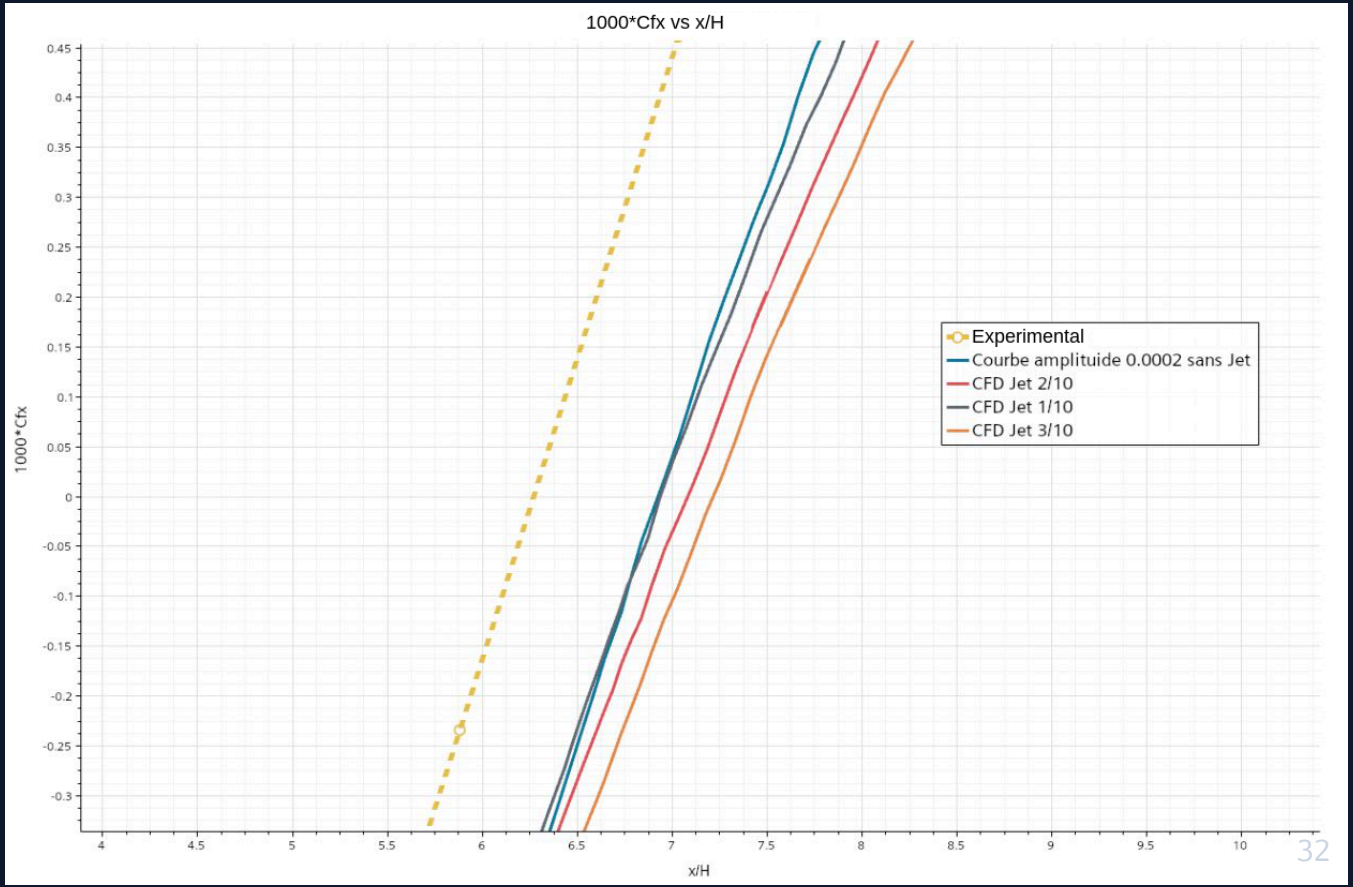
Deviations

No jet: 9.92%

H/10: 9.92%

2H/10: 15.2%

3H/10: 16%



Conclusion & Outlook

Conclusion

Realizable k-ε with a developed inlet profile = best compromise

Outlook

LES for unsteady flow, 3D geometry, heat transfer

Thank you for your attention!

Questions?